

Year 4	9.57%	Year 9	16.54%
Year 5	-40.33%	Year 10	27.37%

1. For Years 6 through 10, the mean absolute deviation (MAD) of the MSCI World Index total returns is *closest* to:

- A. 10.20 percent.
- B. 12.74 percent.
- C. 16.40 percent.

Solution:

A is correct. The MAD is calculated as follows:

Step 1 Sum annual returns and divide by n to find the arithmetic mean ($\bar{X}$) of 16.40 percent.

Step 2 Calculate the absolute value of the difference between each year's return and the mean from Step 1. Sum the results and divide by n to find the MAD:

$$\text{MAD} = \frac{\sum_{i=1}^n |X_i - \bar{X}|}{n},$$

$$\text{MAD} = \frac{50.98\%}{5} = 10.20\%.$$

These calculations are shown in the following table:

Year	Step 1	Step 2
	Return	$ X_i - \bar{X} $
Year 6	30.79%	14.39%
Year 7	12.34%	4.06%
Year 8	-5.02%	21.42%
Year 9	16.54%	0.14%
Year 10	27.37%	10.97%
Sum:	82.02%	50.98%
n :	5	5
$\bar{X}$:	16.40%	10.20%

Annual returns and summary statistics for three funds are listed as follows:

Year	Annual Returns (%)		
	Fund ABC	Fund XYZ	Fund PQR
Year 1	-20.0	-33.0	-14.0
Year 2	23.0	-12.0	-18.0
Year 3	-14.0	-12.0	6.0
Year 4	5.0	-8.0	-2.0
Year 5	-14.0	11.0	3.0